# Thomas Liu

437-232-2668 | thomas.liu.lty@gmail.com | LinkedIn | GitHub

## Education

**University of Waterloo** Sep. 2022 – Present
*Bachelor of Mathematics - Major in Statistics and Computational Mathematics, Computing Minor* Waterloo, ON

- Relevant Courses: Statistical Learning: Advanced Regression, Neural Networks, Computational Inference, Statistical Analysis, Forecasting, Applied Linear Models, Sampling & Experimental Design, Object-Oriented Programming

## Technical Skills

**Languages**: Python, SQL, R, SAS, C, C++, JavaScript, LaTeX, C#, MATLAB, DAX
**Data Tools**: PowerBI, Excel, Snowflake, Git, AWS, Microsoft Azure, Looker/Data Studio, Google Analytics
**Frameworks**: Numpy, Pandas, Matplotlib, TensorFlow, PyTorch, Scikit-learn, XGBoost, PySpark, JSON

## Experience

**Risk Management - Model Validation** Sep. 2026 – Dec. 2026
*TD Bank* *Toronto, ON*

**Data Analyst** May 2026 – Aug. 2026
*Rocket Companies* *Toronto, ON*

- Built a net-new **SQL** and **Python** pipeline to detect loan closing type mismatches, leveraging AI to identify root causes and delivering automated email alerts to directors and VPs for expedited resolution
- Utilized **SQL** to update stored procedures and constructed interactive **PowerBI** dashboard to track AI usage across the mortgage closing stage, driving insights that contributed to lower inbound call volume
- Designed a reusable What/Why/Action **PowerBI** template, enabling stakeholders to gain a high-level report overview within **2 minutes** while providing targeted actions to ensure alignment with company goals

**Data Analyst** Sep. 2025 – Dec. 2025
*Super.com* *Toronto, ON*

- Leveraged **SQL** and **Python** to benchmark QL2 against Luminai for competitor insights, confirming QL2's superior data volume and accuracy with **half the cost**, leading to a strategic supplier transition
- Led dbt model enhancements to incorporate new metasearch data sources, enabling robust cross-channel cMAB analysis and optimized data-driven strategy planning across travel acquisition channels
- Analyzed LLM-based room-matching results using **SQL** and collaborated closely with engineering to tune the model, resolving customer-reported mismatches and increasing overall revenue by **1.5%**

**Data Analyst** Jan. 2025 – Apr. 2025
*Lifestyle Home Products* *Toronto, ON*

- Built comprehensive, dynamic dashboards by integrating **Salesforce** data with **Google Sheets** and **Looker Studio**, enabling real-time insights that supported data-driven decision-making across business teams
- Established a stable **POST API** connection to securely transmit real-time customer data from **Salesforce** to third-party system Bellastone, ensuring reliable and efficient cross-platform data exchange
- Implemented automated data validation workflows within **Salesforce** to ensure up to **98%** data accuracy and completeness, enhancing reporting reliability, sales forecasting, and business intelligence processes

**Data Analyst** May 2024 – Aug. 2024
*Ontario Lottery and Gaming Corporation* *Toronto, ON*

- Conducted post-campaign analysis using **Python** with **Numpy**, **Pandas**, and **Matplotlib**, identifying key audience segments and optimizing marketing strategies to drive a **20%** increase in ROI
- Constructed interactive **PowerBI** dashboards and **SQL** queries to streamline data access for business teams, reducing data request time by up to **90%** and enabling faster, data-driven decision-making
- Collaborated with the Advanced Analytics team to implement the **North Star Metric**, enhancing customer targeting accuracy by at least **10%** for promotions and improving overall campaign performance

**Full Stack Developer** Sep. 2023 – Dec. 2023
*SparkLease Inc.* *Toronto, ON*

- Designed and developed the user profile feature in WeChat Mini App using **React Native**, **XML**, **CSS**, and **JavaScript**, enabling seamless post management and optimizing process speed by **50%**
- Engineered responsive PC and mobile websites with **ASP.NET**, **CSHTML**, **C#**, and **JavaScript**, enhancing user navigation by **30%** through a dynamic filter system and improved account management
- Created a recommendation algorithm using **C#** and **SQL** on Microsoft Azure, increasing data retrieval speed by **30%** and improving recommendation accuracy by **15%**, leading to higher user satisfaction